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Cover deployment system

Abstract

A cover deployment system includes a flexible cover configured to span an open top of a trailer, the trailer having a front end, a rear end, and longitudinally extending side walls extending therebetween, bow assemblies connected to the cover, and a cable system including a pair of cables, an electric actuator, a manual actuator, and a differential. The cables extend longitudinally along the longitudinal side of the trailer. Each cable defines a loop supported by a pulley at the front end of the trailer and a pulley at the rear end of the trailer. The remaining bow assemblies are movably attached to the cables. The electric actuator and the manual actuator are operatively connected to a differential positioned between the electric actuator, the manual actuator, and the pulley. The differential is connected and rotates a pulley to move the cables, thereby moving the cover between closed and open positions.

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Background/Summary

BACKGROUND OF THE INVENTION

(1) This invention relates in general to covering systems used in tractor-trailer hauling applications. In particular, this invention relates to a power actuated top covering system for selectively covering the open top area of a trailer having rigid side walls.

(2) Commercial truck trailers are produced in a variety of forms to support and haul a variety of goods. Flatbed trailers are designed to haul heavy loads, such as coiled steel or larger pieces of equipment, and permit these loads to be placed on the trailer surface easily. Box trailers tend to haul lighter packaged goods that need to be protected from wind loads during transport. Open top, dump trailers configured to haul generally loose and dry goods, such as grain, sand, gravel, and the like, have generally rigid side walls and a rear mounted gate or a bottom-mounted chute or door. The open top permits the goods to be loaded into the trailer. The gate permits goods to be released as the trailer is tilted. The bottom-mounted door permits unloading without the need to tilt the trailer.

(3) Open top dump trailers, because of the loose nature of their cargo, are often covered to prevent road wind from blowing the trailer contents onto the roadway and onto other vehicles following behind. Many states require certain types of loose cargo to be covered to prevent debris from impacting other vehicles. These top coverings may be canvas or rigid panels. Canvas coverings are usually rolled, either to one side or an end of the trailer, to deploy or remove the cover. The rolling mechanism may include a U-shaped “towel” bar that extends along the outer surface of the side walls and pivotally mounts to the lower portion of the trailer. The tarp is rolled onto or off the towel

bar portion that extends across the trailer. Other types of deployment mechanisms may include a hand operated crank that rotates a take-up bar to roll up the tarp covering. These types of deployment devices are prone to damage because they mount on the exterior of the trailer bed or become cumbersome to operate. Thus, it would be desirable to provide a tarp deployment system for an open top trailer that improves deployment and durability.

SUMMARY OF THE INVENTION

(4) This invention relates to a power actuated top covering system for selectively covering the open top area of a trailer having rigid side walls.

(5) In one embodiment, a cover deployment system includes a flexible cover configured to span an open top of a trailer, the trailer having a front end, a rear end, and longitudinally extending side walls extending therebetween, a plurality of bow assemblies connected to the cover, and a cable system including a pair of cables, an electric actuator, a manual actuator, and a differential. The pair of cables extend longitudinally along the longitudinally extending side walls of the trailer adjacent to the open top. Each of the pair of cables define a loop supported by a pulley at the front end of the trailer and a pulley at the rear end of the trailer. One of the plurality of bow assemblies is fixedly attached to the pair of cables at the front end or the rear end of the open top of the trailer. The remaining bow assemblies of the plurality of bow assemblies are movably attached to the pair of cables. The electric actuator and the manual actuator are operatively connected to the differential, and the differential is positioned between the electric actuator, the manual actuator, and the pulley. The differential is connected to at least one of the pullies and is operative to rotate the pulley to move the cables, thereby moving the cover between a closed position and an open position.

(6) In a second embodiment, a cable attachment bracket includes a first end having a substantially planar sliding surface and a bow mounting groove opposite the planar sliding surface. The bow mounting groove is configured for attachment to a distal end of a bow. A second end has an inboard portion and an outboard portion and is configured for sliding attachment to a cable. A first cable groove is formed on an outwardly facing surface of the inboard portion, and a second cable groove is formed on an inwardly facing surface of the outboard portion. Prior to the attachment of the cable attachment bracket to the cable, the inboard portion and the outboard portion of the second end have a V-shaped traverse section and define an open position.

(7) In a third embodiment, a bow assembly includes a pair of cable attachment brackets, each cable attachment bracket having a first end having a substantially planar sliding surface and a bow mounting groove opposite the planar sliding surface. The bow mounting groove is configured for attachment to a distal end of a bow, and a second end has an inboard portion and an outboard portion and is configured for sliding attachment to the cable. A first cable groove is formed on an outwardly facing surface of the outboard portion, and a second cable groove is formed on an inwardly facing surface of the outboard portion. Prior to attachment of the cable attachment bracket to the cable, the inboard portion and the outboard portion of the second end have a V-shaped traverse section and define an open position. Each distal end of the bow is attached to one of the pair of cable attachment brackets. When the cable is positioned within the first and second cable grooves, and when the inboard portion and the outboard portion of the second end are urged toward each other, the inboard portion and the outboard portion of the second end define a closed position wherein the first and second cable grooves define a cable channel, and the cable is slidably mounted within the cable channel.

(8) Various aspects of this invention will become apparent to those skilled in the art from the following detailed description of the preferred embodiment, when read in light of the accompanying drawings.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

(1) FIG. **1** is a perspective view of a first side of a trailer with a first embodiment of a cover deployment system according to the invention.

(2) FIG. **2** is an alternate perspective view of the trailer and a portion of the cover deployment system illustrated in FIG. **1** showing a top side and a second side of the trailer with a portion of the cover removed.

(3) FIG. **3** is a perspective view of a front end of the trailer showing a protective plate and a portion of the cover deployment system illustrated in FIG. **1**.

(4) FIG. **4** is an enlarged perspective view of the electric actuator illustrated in FIG. **3**.

(5) FIG. **5**A is a perspective view of an open differential.

(6) FIG. **5**B is a perspective view of a Torsen differential.

(7) FIG. **6** is a perspective view of a first embodiment of a manual actuator.

(8) FIG. **7** is perspective view of the first embodiment of the manual actuator illustrated in FIG. **6** with the handle and the fourth sprocket pulley removed.

(9) FIG. **8** is a perspective view of a second embodiment of the manual actuator.

(10) FIG. **9** is a perspective view of a cable attachment bracket.

(11) FIG. **10** is a perspective view of the cable attachment bracket illustrated in FIG. **9** shown attached to a first cable.

(12) FIG. **11** is a perspective view of the cable attachment bracket attached to the trailer by a sliding bracket assembly.

(13) FIG. **12** is an enlarged perspective view of bow assemblies connected to the first cable showing one of the bow assemblies attached to the trailer by the sliding bracket assembly.

(14) FIG. **13** is an enlarged perspective view of a rear bow assembly attached to the trailer by a rear sliding bracket assembly and a cable tensioner.

(15) FIG. **14** is an alternate enlarged perspective view of the rear bow assembly illustrated in FIG. **13** showing the rear sliding bracket assembly removed.

(16) FIG. **15** is a second alternate enlarged perspective view of the rear bow assembly and the cable tensioner with a tension pulley cover removed.

(17) FIG. **16** is an enlarged perspective view of the second side of the trailer showing a second pulley and a second cable.

(18) FIG. **17** is a perspective view of an alternate embodiment of the cover including a plurality of pockets.

(19) FIG. **18** is a perspective view of an alternate embodiment of the cable tensioner.

(20) FIG. **19** is a plan view of the cable tensioner illustrated in FIG. **18** show in an untensioned state.

(21) FIG. **20** is an alternate plan view of the cable tensioner illustrated in FIGS. **18** and **19** show in a tensioned state.

DETAILED DESCRIPTION OF THE PREFERRED EMBODIMENT

(22) Throughout this disclosure, various publications, patents, and published patent specifications are referenced by an identifying citation. The disclosures of these publications, patents, and published patent specifications are hereby incorporated by reference into the present disclosure in their entirety to more fully describe the state of the art to which this invention pertains.

(23) The term “chain” herein is used to describe a loop structure in which can be used to operatively connect two pulleys. The term “chain” can include, but is not limited to, a belt, a chain, and a cable.

(24) Referring now to FIG. **1**, a trailer **10** is shown with a first embodiment of a cover deployment system **12** according to the present invention. The cover deployment system **12** is configured to retract and extend a cover **14** over the trailer **10**. The illustrated trailer **10** has a front end **16** (the left end when viewing FIG. **1**) configured for attachment to a prime mover, such as a tractor (not

shown), a rear end **18** (the right end when viewing FIG. **1**), a first side **20**, a second side **22**, best shown in FIG. **2**, a top side **24**, and a bottom side **26**. The trailer **10** may be any type of cargo containment vessel having a generally open top surface, such as a dump bed, box container, grain hauler, and the like. The front end **16** includes a protective plate **28**, described below, extending upwardly from the top side **24**. Additionally, a ledge **29** extends outwardly from the bottom side **26** of the front end **16**.

(25) The cover deployment system **12** includes the cover **14**, and also includes a cable system **30**, and a plurality of bow assemblies **32**. In the figures, the cover **14** is shown as being formed from a soft material, such as plastic, canvas, woven fabric, and other flexible material. Alternatively, the cover **14** may be configured as a hard covering structure, such as a segmented series of metal or polymer panels. An alternate embodiment of the cover **14′** is shown in FIG. **17** and includes a plurality of sleeves or pockets **15** into which each bow assembly **32** may be inserted and maintained.

(26) The cable system **30** includes an electric actuator **34** consisting of a motor **90**, a gearbox **92**, and an output shaft **98**, a manual actuator, such as the manual actuator **36**, best shown in FIG. **6**, or the manual actuator **38**, shown in FIG. **8**, a bar **40**, the chains **82** and **88**, a first cable **72**, a first pulley **70**, sprocket pullies **80**, **84**, **86**, and **202** or **310**, and may include a cable tensioner **42**, best shown in FIG. **13**. Each of the sprocket pullies **80**, **84**, **86**, and **202** or **310** can have sprockets **81**, best shown in FIGS. **4** and **6-8**, configured to move the chain **82** or **88**. For example, in the embodiment of the cable system **40** shown in FIG. **1**, the first pulley **70** is operatively connected to a tension pulley **43**, best shown in FIGS. **13-15**, of the cable tensioner **42** by the first cable **72**. A first sprocket pulley **80** is operatively connected to the manual actuator **36** by the first chain **82** (see FIG. **1**), and a second sprocket pulley **84** is operatively connected to a third sprocket pulley **86** by the second chain **88** (see FIG. **4**). These elements of the cable system **30** will be explained in greater detail below.

(27) As best shown in FIGS. **9** through **12**, the illustrated bow assemblies **32** include arcuate bows **44**, the distal ends of which are attached to cable attachment brackets **46**, configured for sliding engagement with the top side **24** of the trailer **10**. As best shown in FIG. **9**, each cable attachment bracket **46** includes a first end **44**A configured for attachment to a distal end of the bow **44** and a second end **44**B configured for sliding attachment to one of two longitudinally extending cables **72** and **78**, described below.

(28) The first end **44**A of the bracket **46** includes a substantially planar sliding surface **47** and a bow mounting groove **48** formed in an outwardly facing surface thereof (the upwardly facing surface when viewing FIG. **9**). A plurality of bolt holes **50** are formed through opposing side walls of the bow mounting groove **48**. As best shown in FIG. **11**, the distal end of the bow **44** is attached within the bow mounting groove **48** by two or more bolts **52**.

(29) The second end **44**B of the bracket **46** includes a first, inboard portion **54**A and a second, outboard portion **54**B. A first cable groove **56**A is formed on an outwardly facing surface of the inboard portion **54**A and a second cable groove **56**B is formed on an inwardly facing surface of the outboard portion **54**B. Prior to installation, the inboard portion **54**A and the outboard portion **54**B of the second end **44**B have a V-shaped transverse section and define a cable mounting slot **58**. When installed, the first cable groove **56**A and the second cable groove **56**B combine to define a cable channel **60**.

(30) A first bolt hole **62**A is formed through the second end **44**B inboard of the cable channel **60**, and a second bolt hole **62**B is formed through the second end **44**B outboard of the cable channel **60**. When the second end **44**B is installed about one of the two longitudinally extending cables **72** and **78**, as shown in FIGS. **10** and **11**, bolts **52** and attached nuts **53** may be used to urge the inboard portion **54**A and the outboard portion **54**B of the second end **44**B of the bracket **46** together.

(31) As shown in FIGS. **11** and **12**, the bow assemblies **32** that are configured for sliding movement are further attached to the trailer **10** by a plurality of spaced apart sliding bracket assemblies **64**.

Each sliding bracket assembly **64** includes a first bracket portion **66** attached to either the first side **20** or the second side **22** of the trailer **10**. Each first bracket portion **66** has an L-shaped transverse section and includes a first planar wall configured for attachment to the trailer **10**, and a second planar wall extending outwardly from the trailer **10**. The lower surface of the second planar wall (downwardly facing when viewing FIG. **11**) defines a first sliding surface **66**A.

(32) The sliding bracket assembly **64** also includes an L-shaped second bracket portion **68**. A first end of the second bracket portion **68** is attached to the cable attachment bracket **46** via the bolts **52** extending through the first and second bolt holes **62**A and **62**B, and a second end of the second bracket portion **68** extends inwardly toward the trailer **10**. The upper surface of the second end of the second bracket portion **68** (upwardly facing when viewing FIG. **11**) defines a second sliding surface **64**A, configured for sliding engagement with the first sliding surface **66**A. Additionally and advantageously, each cable attachment bracket **46** may be attached to the first cable **72** with only one bolt **52** that extends through the second blot hole **62**B.

(33) Referring now to FIGS. **13** through **15**, a rear bow assembly **400** is shown. The rear bow assembly **400** uses the previously described bow assemblies **32**. The illustrated rear bow assembly **400** includes two bow assemblies **32**, however, any number of bow assemblies **32** including, but not limited to, one, two, three, or four may be used. The rear bow assembly **400** may be on the first side **20** and/or the second side **22** of the trailer **10**. In the illustrated rear bow assembly **400**, the two bow assemblies **32** are locked into position on the first cable **72** by means of locking members **402** on each side of the rear bow assembly **400**. The rear bow assembly **400** is attached to the trailer **10** by a rear sliding bracket assembly **404** best shown in FIGS. **13** and **15**.

(34) The two bow assemblies **32** are fixedly attached to each other by an L-shaped rear first bracket portion **406**. A first end of the rear first bracket portion **406** is attached to the cable attachment brackets **46** via the bolts **52** extending through the first and second bolt holes **62**A and **62**B of the cable attachment brackets **46**, and a second end of the rear first bracket portion **406** extends inwardly toward the trailer **10**, defining a first sliding surface **406**A.

(35) The rear sliding bracket assembly **404** includes a rear second bracket portion **408** attached the trailer **10**. The rear second bracket portion **408** has an L-shaped traverse section and includes a first planar wall configured for attachment to the trailer **10**, and a second planar wall extending outwardly from the trailer **10**. The lower surface of the second planar wall (downwardly facing when viewing FIG. **15**) defines a second sliding surface **408**A, configured for sliding engagement with the first sliding surface **406**A.

(36) Referring now to FIGS. **13** and **14**, the cable tensioner **42** is shown. The illustrated cable tensioner **42** is attached to the first side **20** of the trailer **10** by a fastener **43**B, however, the cable tensioner **42** may also be attached to the second side **22**, or the cable tensioner **42** may be attached to both the first and second sides **20** and **22**. The tension pulley **43** of the cable tensioner **42** is rotatably attached to the first side **20** by the fastener **43**B. A tension mechanism **42**A is fixedly attached to the tension pulley **43**. The tension mechanism **42**A has a U-shaped bracket **42**B that surrounds a portion of the tension pulley **43** and is fixedly attached by the fastener **43**B. A threaded opening is formed on a closed end of the bracket **42**B into which a tension bolt **42**C is threaded. The tension bolt **42**C, when rotated clockwise, is configured to apply a force on the tension pulley **43** to apply tension to the first cable **72**. The tension bolt **42**C has a tension nut **42**D thereon configured to prevent the bolt **42**C from unscrewing or rotating counterclockwise. As shown in FIG. **13**, the cable tensioner **42** may include a tension pulley cover **43**A attached via the fastener **43**B and configured to protect the tension pulley **43** from damage, such as from airborne objects.

(37) Referring now to FIGS. **18** through **20**, an alternative embodiment of the cable tensioner is shown at **430**. The cable tensioner **430** includes a bracket assembly **432** having a mounting portion **434**, as guide portion **436**, and pulley cover **438**. The mounting portion **434** has a L-shaped transverse section and includes a longitudinally extending first side **440** and longitudinally extending second side **442**. The first side **440** is configured to be attached to a side of the trailer **10**

by bolts (not shown) via mounting holes **444**. The second side **442** includes two traverse slots **446** and two openings **448**.
(38) The guide portion **436** has a U-shaped transverse section and includes a longitudinally extending base **450** and two longitudinally extending side walls **452**A and **452**B. The side wall **452**A is positioned adjacent the first side **440** of the mounting portion **434**. The side wall **452**B includes a first alignment window **454**, the purpose for which will be explained below. The base **450** is attached to the second side **442** by fasteners, such as bolts (not shown) and nuts **458** that extend through the slots **446**. The base **450** also includes two longitudinally extending slots **451** aligned within the two openings **448**. A spring retainer **470** has and L-shaped transverse section and includes a first leg **472** and a second leg **474**. The first leg **472** is attached to first end of the guide portion **436** (the left end when viewing FIGS. **19** and **20**) and includes a pin hole **476** formed therethrough. A spring mounting pin (not shown) extends through the pin hole **476**.
(39) The pulley cover **438** includes a side wall **460** having a forwardly extending spring arm **462** (to the left when viewing FIGS. **19** and **20**) having a transversely extending engagement surface **462**A. The pulley cover **446** further includes a longitudinally extending first or upper flange **464** and a parallel, longitudinally extending second or lower flange **466**. The lower flange **466** is movably mounted to the base **450** by bushings **468** that extend from the lower flange **466**, through the slots **451**, and into the openings **448**. A second alignment window **467** is formed in the side wall **460**. A coil spring **478** is positioned between the first leg **472** of the spring retainer **470** and the spring arm **462** and secured in place by the spring mounting pin (not shown). Like the cable tensioner **42**, the cable tensioner **430** includes the tension mechanism **42**A fixedly attached to the tension pulley **43**.
(40) In FIG. **19**, the cable tensioner **430** is shown in an untensioned state wherein the spring **478** is not compressed and the first and second alignment windows **454** and **467**, respectively, are not aligned.
(41) In FIGS. **18** and **20**, the cable tensioner **430** is shown in a tensioned state wherein the spring **478** is compressed and the first and second alignment windows **454** and **467**, respectively, are aligned.
(42) Advantageously, the cable tensioner **430** gives an operator a simple and effective means of providing a known tension on the cable, such as the first cable **72**. For example, an operator must only adjust the tension mechanism **42**A until the pulley cover **446** moves toward the spring **478** (leftward when viewing FIGS. **19** and **20**), thus compressing the spring **478**. Once the operator observes that the first and second alignment windows **454** and **467** are aligned, the operator will know that the tension on the cable has reached a pre-determined and pre-set value. For example, the illustrated spring **478** may be compressed until a 400 ft-lb tension is achieved in the first cable **72**.
(43) In the illustrated embodiments, a 400 ft-lb cable tension has been shown to keep the bows **44** in line and prevent binding of the cover **14** during operation. It will be understood however, that an optimum cable tension will vary based on the size and configuration of the cover **14** and the cover deployment system **12**.
(44) FIG. **2** is an alternate view of the trailer **10** and the cover deployment system **12**, and shows the top side **24**, the second side **22**, and the rear end **18** with the cover **14** deployed with a section of the cover **14** removed. FIG. **2** also shows the plurality of bow assemblies **32** that span the top side **24**, and a second pulley **74** and a third pulley **76** operatively connected by a second cable **78**.
(45) FIG. **3** is an enlarged view of the front end **16** showing the protective plate **28**, the bar **40**, and the electric actuator **34**. The protective plate **28** may be positioned using a series of plate slits **28**A and fasteners, such as plate bolts **28**B. Alternatively, any desired fastener may be used, such as for example, screws, studs, and the like. The illustrated protective plate **28** may be positioned by loosening the plate bolts **28**B, sliding the protective plate **28** to a desired position, and tightening the plate bolts **28**B to retain the protective plate **28** in the desired position. Alternatively, any

desired means of attachment that allows the protective plate **28** to be selectively moved to, and fixed in, a desired position may be used. Additionally, the protective plate **28** may also be fixedly attached to the trailer **10**. The protective plate **28** is configured to protect the cover **14** from airborne hazards when the trailer **10** is in motion, both when the cover **14** is compressed forward, i.e., toward the front end **16**, and when the cover **14** is extended, as shown in FIGS. **1** and **2**.

(46) Referring again to FIG. **3**, the cable system **30** includes the first pulley **70** and the second pulley **74**, both fixedly attached to the bar **40**. The bar **40** may attached to the trailer with at least one bar holder **40**A. The bar holder **40**A is secured to the trailer **10** by a plurality of fasteners, such as the bolts **52**. In FIG. **3**, four bar holders **40**A are illustrated, however, any number of bar holders **40**A may be provided. The bar **40** may include a generally cylindrical, bar adjustment mechanism **40**C disposed around a portion of the bar **40** as shown in FIG. **3**. The bar adjustment mechanism **40**C is configured to adjust a bar length **40**B to accommodate trailers **10** with different trailer widths **10**A. The bar length **40**B may be adjusted by removing or loosening a fastener from the bar adjustment mechanism **40**C, adjusting the length of the bar **40**, and replacing or tightening the fastener. The bar adjustment mechanism **40**C may be formed from any desired rigid material known including, but not limited to, steel and iron. The bar adjustment mechanism **40**C may have any desired shape that does not inhibit the rotation of the bar **40**. However, the bar **40** does not need a bar adjustment mechanism **40**C to function properly.

(47) Referring now to FIGS. **4**, **5**A, and **5**B, the electric actuator **34** and two possible types of differentials are shown. In the illustrated electric actuator **34**, the motor **90** is operatively connected to the gearbox **92** that is fixedly attached to a portion of the trailer **10** using a gearbox bracket **94**. The gearbox bracket **94** may be mounted to a gearbox mounting ledge **96** of the trailer **10** by any desired fasteners, such as bolts, rivets, and the like. The output shaft **98** extends from the gearbox **92** and is operatively connected to an input shaft **100** by a coupler **102**. The coupler **102** is configured to connect and disconnect the motor **90** and the gearbox **92** from a differential **104**.

(48) The differential **104** may be any type of differential such as an open differential, best shown in FIG. **5**A, or a Torsen differential **107**, best shown in FIG. **5**B. As shown in FIG. **4**, the differential **104** is an open differential. The input shaft **100** is part of the differential **104** and has a first gear **106** that is engaged with the differential gears **108** and a second gear **110** that is also engaged with the differential gears **108**. The second gear **110** has a differential output shaft **112** that is fixedly attached to the second sprocket pulley **84**. The first sprocket pulley **80** is fixedly attached to the differential **104** and configured to be operatively attached to the manual actuator **36** or **38**. The third sprocket pulley **86** is connected to the bar **40** by a breakaway clutch (not shown) and is operatively connected to the second sprocket pulley **84** by a second chain **88**. The breakaway clutch is mounted within, or adjacent to, the third sprocket pulley **86**, and is configured to stop movement of the cover **14** via spring washers that will engage the breakaway clutch. Thus, the breakaway clutch is operatively connected between the bar and the third sprocket pulley **86**. The breakaway clutch, when engaged, will also reduce stress on the motor **90**. In the illustrated embodiments, the breakaway clutch is configured to activate in any overload situation, including obstructions and end of travel limits. For example, depending on the on the size and configuration of the cover deployment system **12**, the vehicle to which it is mounted, and the performance characteristics desired by the operator, the breakaway clutch can be set to operate, i.e., to release, within the range of from about 0 in-lbs to about 3,000 in-lbs of torque.

(49) The Torsen Differential **107** illustrated in FIG. **5**B is conventional in the art and has a first shaft **107**A that is operatively attached to a series of gears **107**B contained within a housing **107**C. The gears **107**B rotate due to rotational torque provided by the first shaft **107**A. The series of gears **107**B are also operatively attached to a second shaft **107**D that is configured to transfer torque to the second sprocket pulley **84**.

(50) Referring now to FIGS. **6** and **7**, the first embodiment of the manual actuator **36** is shown. The fourth sprocket pulley **202** is rotatably attached to the ledge **29** by a bracket **200**. The fourth

sprocket pulley **202** and the first sprocket pulley **80**, best shown in FIG. **4**, are operatively connected by the first chain **82**. The manual actuator **36** is attached to the ledge of the trailer **10** by a crank retention pin **204** that extends outwardly from the ledge **29** and through the fourth sprocket pulley **202**. A crank **206** is fixedly attached to the crank retention pin **204** and may extend generally perpendicularly from a longitudinally extending axis of the crank retention pin **204**. A crank handle **208** extends outwardly (generally perpendicularly from the crank **206** when viewing FIG. **6**) from a distal end of the crank **206** for user operation.

(51) A crank lock **210** is positioned between the fourth sprocket pulley **202** and the ledge **29**. The crank lock **210** has a first, generally C-shaped, brake member **212** and a second, generally C-shaped, brake member **214** that are fixedly attached to the ledge **29** by bolts **52** at a point distal from a crank lock handle **216**. The first brake member **212** and the second brake member **214** are operatively connected to the crank lock handle **216** by a screw **218** that extends between the first brake member **212** and the second brake member **214**.

(52) The bolts **52** function to secure the first brake member **212** and the second brake member **214** in position, but also allow the brake members **212** and **214** to be adjusted closer together at a point distal from the bolts **52**. This adjustment may be achieved by rotating the crank lock handle **216** to adjust the length of a portion of the brake screw **218**. The length of the portion of the brake screw **218** indicates how close the first brake member **212** is to the second brake member **214**. When the brake members **212** and **214** are adjusted to be close together, a forced is applied to the fourth sprocket pulley **202**, which prevents the fourth sprocket pulley **202** from rotating. Thus, when the brake members **212** and **214** are adjusted closer together the manual actuator **36** is in a locked position. However, when the brake members **212** and **214** are adjusted to be further apart, the manual actuator **36** is in an unlocked position.

(53) Referring now to FIG. **8**, the second embodiment of the manual actuator **38** is shown. The manual actuator **38** includes a crank mount **300** configured for attachment to the ledge **29** of the trailer **10**. The crank mount **300** includes two co-planar mounting flanges **300**A extending outwardly from an intermediate portion **300**C and defining distal ends thereof. The flanges **300**A have mounting apertures **300**B formed therethrough. The crank mount **300** may be attached to the ledge **29** of the trailer **10** by one or more fasteners (not shown) that extend through the mounting apertures **300**B.

(54) The intermediate portion **300**C has a generally circular shape, has a plurality of radially extending crank slots **304** formed in a peripheral edge thereof, and is connected to the flanges **300**A via connecting portions **300**D. The connecting portions **300**D extend outwardly from the flanges **300**A such that a plane of the intermediate portion **300**C is spaced a distance apart from the ledge **29** when the crank mount **300** is mounted to the trailer **10**. A second embodiment of the fourth sprocket pulley **310** is rotatably attached to the intermediate portion **300**C of the crank mount **300**. The fourth sprocket pulley **310** is operatively connected to the first sprocket pulley **80**, best shown in FIG. **4**, by the first chain **82**.

(55) A crank bracket **312** includes a base **314** having a first wing **316** and a second wing **318** extending outwardly therefrom. The crank bracket **312** is fixedly attached to the crank mount **300** by a bolt **314**A. The wings **316** and **318** have holes therethrough through which a crank **306** may extend. The illustrated crank **306** includes a first portion **306**A, a second portion **306**B, extending at about a 90-degree angle from the first portion **306**A, and a third portion **306**C, extending at about a 90-degree angle from the second portion **306**B. An elbow **308** is defined at the bend between the second portion **306**B and the third portion **306**C. A handle **320** extends at about a 90-degree angle from the third portion **306**C and is configured for user operation of the manual actuator **38**. The first portion **306**A may be retained within the wing **318** by a fastener **307**.

(56) The second portion **306**B of the crank **306** near the elbow **308** is configured to be seated within one of the radially extending crank slots **304**, and to be moved as desired to any of the plurality of radially extending crank slots **304**. Thus, the manual actuator **38** is configured to be

moved between a desired one of a plurality of locked positions, wherein the second portion **306**B of the crank **306** is secured within one of the crank slots **304** and an unlocked position wherein the second portion **306**B of the crank **306** is not seated or secured within one of the radially extending crank slots **304**.

(57) In operation, the cover deployment system **12** may use a control panel (not shown), which may be cab-mounted, to activate the motor **90**. The motor **90** provides rotary torque to the gearbox **92** which in turn reduces speed and increases the torque that is applied to the output shaft **98**. The output shaft **98** then provides torque through the differential **104** causing the second sprocket pulley to rotate **84**. When the second sprocket pulley **84** rotates, it moves the second chain **88** and causing the third sprocket pulley **86** rotate. The third sprocket pulley **86** then rotates the bar **40** which in turn rotates the first pulley **70** and the second pulley **74**. When the first pulley **70** and the second pulley rotate **74**, the first cable **72** and the second cable **78** move, respectively. The cables **72** and **78** then move the rear bow assembly **400** along the cables **72** and **78** extending or retracting the cover **14**. The cover **14** is configured such that when the cover **14** retracts towards the front end **16** of the trailer **10**, the cover material folds like an accordion. When the cover **14** extends or retracts, the pockets **15** in the cover **14** slide the remaining bow assemblies **32** along the sliding bracket assemblies **64**.

(58) If manual actuation is desired, the manual actuator **36** or **38** may be operated by rotating the crank **206** or **306** which rotates the fourth sprocket pulley **202** or **310**. The fourth sprocket pulley **202** or **310** then moves the first chain **82**, which rotates the first sprocket pulley **80**. When the first sprocket pulley rotates **80**, the differential **104** rotates. The differential **104** rotation causes the differential output shaft **112** rotate and further causes the second sprocket pulley **84** rotate. When the second sprocket pulley **84** rotates, it moves the second chain **88**. This causes the third sprocket pulley **86** rotate. The third sprocket pulley **86** then rotates the bar **40** which in turn rotates the first pulley **70** and the second pulley **74**. When the first pulley **70** and the second pulley **74** rotate, the first cable **72** and the second cable **78** move, respectively. The cables **72** and **78** then move the rear bow assembly **400** along the cables **72** and **78** extending or retracting the cover **14**. The cover **14** is configured such that when the cover **14** retracts towards the front end **16** of the trailer **10**, the cover material folds like an accordion. When the cover **14** extends or retracts, the pockets **15** in the cover **14** slide the remaining bow assemblies along the sliding bracket assemblies **64**.

(59) Although the cover deployment system **12** disclosed herein has been described as being used on a trailer with an open top, it will be understood that the various embodiments of the cover deployment system **12** may be used with covers for any desired container or structure having an open or closed top.

(60) The principle and mode of operation of this invention have been explained and illustrated in its preferred embodiment. However, it must be understood that this invention may be practiced otherwise than as specifically explained and illustrated without departing from its spirit or scope.

Claims

1. A cover deployment system comprising: a flexible cover configured to span an open top of a trailer, the trailer having a front end, a rear end, and longitudinally extending side walls extending therebetween; a plurality of bow assemblies connected to the cover; and a cable system including a pair of cables, an electric actuator, a manual actuator, and a differential; wherein the pair of cables extend longitudinally along the longitudinally extending side walls of the trailer adjacent the open top, each of the pair of cables defining a loop supported by a pair of pullies, each pair of pullies having a first pulley at the front end of the trailer and a second pulley at the rear end of the trailer; wherein one of the plurality of bow assemblies is fixedly attached to the pair of cables at one of the front end and the rear end of the open top of the trailer; wherein the remaining bow assemblies of the plurality of bow assemblies are movably attached to the pair of cables; wherein the electric

actuator and the manual actuator are operatively connected to the differential, the differential positioned between the electric actuator, the manual actuator, and one of the first pullies; and wherein the differential is connected to at least one of the first pullies and is operative to rotate the connected one of the first pullies to move the pair of cables, thereby moving the cover between a closed position and an open position.

2. The cover deployment system of claim 1, wherein the electric actuator and the manual actuator are configured to work independently of each other, such that the cover deployment system is movable between a first operating position wherein the cover is moved by the electric actuator, and a second operating position wherein the cover is moved by the manual actuator.

3. The cover deployment system of claim 1, wherein the differential is a limited slip differential.

4. The cover deployment system of claim 1, wherein the differential is a Torsen differential.

5. The cover deployment system of claim 1, wherein the electric actuator comprises a motor, a gearbox, and an output shaft.

6. The cover deployment system of claim 5, wherein the output shaft is connected to the differential by a coupler that connects a portion of the output shaft to an input shaft of the differential.

7. The cover deployment system of claim 1, wherein the cover includes a pocket configured to receive and maintain one of the pluralities of bow assemblies.

8. The cover deployment system of claim 1, wherein the manual actuator includes a fourth sprocket pulley and wherein a first sprocket pulley is attached to the differential.

9. The cover deployment system of claim 8, wherein the differential and the manual actuator are operatively connected by a first chain that defines a loop that is supported by the fourth sprocket pulley and the first sprocket pulley; and wherein sprockets on the first sprocket pulley and the fourth sprocket pulley are configured to move the chain.

10. The cover deployment system of claim 1, wherein the differential has a differential output shaft fixedly attached to a second sprocket pulley; wherein the second sprocket pulley is operatively attached to a third sprocket pulley by a second chain that defines a loop in which the second chain is configured to rotate the third sprocket pulley; wherein the third sprocket pulley is connected to a bar by a breakaway clutch configured to reduce stress on a motor of the electric actuator when the motor is engaged; and wherein the at least one of the pullies is fixedly attached to the bar.

11. The cover deployment system of claim 1, wherein at least one of second pullies at the rear end of the trailer is a tension pulley configured to apply tension to one of the pair of cables.

12. The cover deployment system of claim 1, further comprising a sliding bracket assembly.

13. The cover deployment system of claim 12, wherein the sliding bracket assembly includes a first bracket portion attached to a side of the trailer and a second bracket portion attached to a cable attachment bracket.

14. The cover deployment system of claim 13, wherein the cable attachment bracket includes a first end having a substantially planar sliding surface and a bow mounting groove opposite the planar sliding surface, the bow mounting groove configured for attachment to a distal end of a bow of one of the bow assemblies, and a second end having an inboard portion and an outboard portion and configured for sliding attachment to one of the pairs of cables; wherein a first cable groove is formed on an outwardly facing surface of the inboard portion and a second cable groove is formed on an inwardly facing surface of the outboard portion; and wherein prior to attachment of the cable attachment bracket to the one of the pairs of cables, the inboard portion and the outboard portion of the second end have a V-shaped traverse section and define an open position.

15. The cover deployment system of claim 14, wherein when the one of the pairs of cables is positioned within the first and second cable grooves, and wherein the inboard portion and the outboard portion of the second end are urged toward each other, the inboard portion and the outboard portion of the second end define a closed position such that the first and second cable grooves define a cable channel.

16. The cover deployment system of claim 15, wherein the one of the pairs of cables is slidably

mounted within the cable channel.

17. The cover deployment system of claim 1, further including a cable tensioner mounted to the trailer and configured to apply tension to at least one of the pair of cables.

18. The cover deployment system of claim 17, wherein the cable tensioner includes: a tension pulley; a bracket assembly; a tension mechanism mounted to the bracket assembly; and a spring within the bracket assembly and configured to apply the tension to the at least one of the pair of cables.